Project Design Phase-II
Technology Stack (Architecture & Stack)

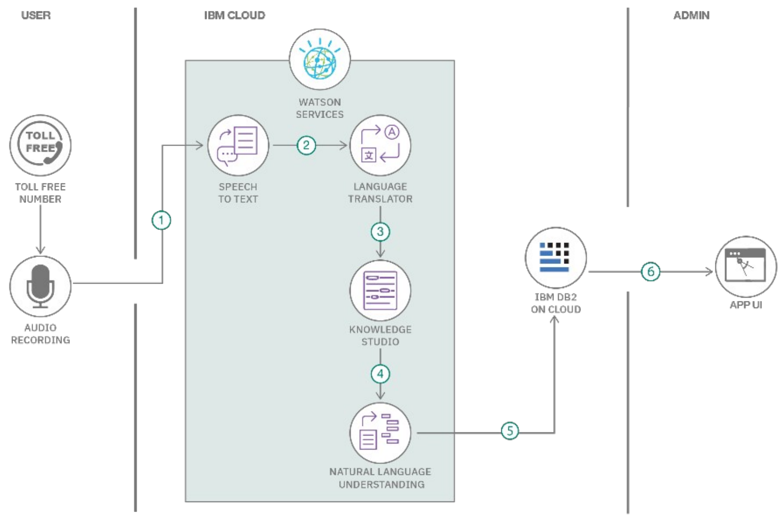
Date	21 June 2026
Team ID	24EG109A55 (Turlapati Lokesh)
Project Name	SHOPEZ - Online Shopping Application
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1. 1.	User Interface	How the user interacts with the application - Web UI (responsive)	React JS, Tailwind CSS / CSS3, Axios
2. 2.	Application Logic-1	Handles authentication & authorization	Node.js, Express.js, JWT, bcrypt
3. 3.	Application Logic-2	Handles product catalog & cart/order logic	Node.js, Express.js REST APIs
4. 4.	Application Logic-3	Handles admin/seller product & order management	Node.js, Express.js, Mongoose
5. 5.	Database	Stores users, products, cart and order data	MongoDB (NoSQL)
6. 6.	Cloud Database	Database hosted on cloud for accessibility	MongoDB Atlas
7. 7.	File Storage	Storage for product images	Cloudinary / Local filesystem
8. 8.	External API-1	Not used in current scope	-
9. 9.	External API-2	Not used in current scope	-
10. 10	Machine Learning Model	Not used in current scope	-

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11. 1 1 .	Infrastructure (Server / Cloud)	Application deployment on cloud Backend: Render Frontend: Vercel	Render, Vercel

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1. 1.	Open-Source Frameworks	React JS, Express.js, Node.js, Mongoose	MERN Stack
2. 2.	Security Implementations	Password hashing, JWT-based auth, protected routes, input validation	bcrypt, JWT, express-validator
3. 3.	Scalable Architecture	3-tier architecture (React frontend, Express/Node backend, MongoDB database) with separate REST APIs per module	MERN 3-tier architecture
4. 4.	Availability	Cloud-hosted backend and frontend with environment-based configuration for uptime	Render + Vercel
5. 5.	Performance	Indexed MongoDB queries, pagination on product listing, lazy loading of images on frontend	MongoDB Indexing, React lazy loading

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d